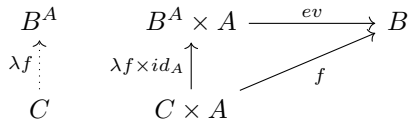
Chapter 3

Cartesian Closed Categories

3.1 Exponentials

Definition 3.1.1 (Exponential)**.** Suppose that $\mathcal{C}$ is a category with finite products. An *exponential* of B by A in $\mathcal{C}$ is specified by a pair that consists of:

- An object B^A
- An *evaluation* morphism $ev_{A,B} : B^A \times A \longrightarrow B$ such that for any $C \in ob\mathcal{C}$ and $f : C \times A \longrightarrow B$, there is a unique morphism $\lambda f : C \longrightarrow B^A$ (called the *transpose* or the *exponential mate* of f) such that $f = ev \circ (\lambda f \times id_A)$, i.e. the following diagram commutes.



Definition 3.1.2 (Cartesian closed category)**.** A Cartesian closed category is a category with finite products such that any pair of objects A and B has an exponential.

Remark 3.1.3. Each $f : C \times A \longrightarrow B$ uniquely determines its transpose $\lambda f : C \longrightarrow B^A$, by definition, but it is also determined by its transpose, since $f = ev \circ (\lambda f \times id_A)$.

Remark 3.1.4. Every morphism $g : C \longrightarrow B^A$ is the transpose of the morphism $\overline{g} \stackrel{def}{=} ev \circ (g \times id_A) : C \times A \longrightarrow B$. Note that we also call $\overline{g}$ the transpose of g, while we call λf the transpose of f.

Exercise 3.1.5. Given a Cartesian closed category with morphisms $f : C \longrightarrow B$ and $h : B \times A \longrightarrow T$, show that the transpose of $h \circ (f \times id_A)$ is $\lambda h \circ f$. Conclude that for any $k : B \longrightarrow T^A$ the transpose of $k \circ f$ is $\overline{k} \circ (f \times id_A)$.

Exercise 3.1.6. Show that, if (B^A, ev) is an exponential of B by A, then $ev = \overline{id_{B^A}}$.

Exercise 3.1.7. Show that, if (I, ev_i) and (J, ev_j) are both exponentials of B by A, then $I \cong J$.

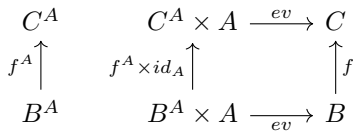
Exercise 3.1.8. Show that the following properties hold in a Cartesian closed category.

1. $(B + C) \times A \cong (B \times A) + (C \times A)$
2. $(B \times C)^A \cong B^A \times C^A$
3. $C^{(A \times B)} \cong (C^B)^A \cong (C^A)^B$
4. $B^{(A+C)} \cong B^A \times B^C$

3.2 Internalizing Composition

Definition 3.2.1. In a Cartesian closed category, the morphism $f^A : B^A \longrightarrow C^A$, the *exponentiation of a morphism* $f : B \longrightarrow C$, is defined to be the transpose of $f \circ ev : B^A \times A \longrightarrow C$.

Exercise 3.2.2. Show that f^A is a unique morphism that makes the following square commute:



Exercise 3.2.3. Show that $(id_B)^A = id_{B^A}$ and $(g \circ f)^A = g^A \circ f^A$.

Exercise 3.2.4. Given a Cartesian closed category with morphisms $f : C \longrightarrow T$ and $h : B \longrightarrow C^A$, show that the transpose of $f^A \circ h$ is $f \circ \overline{h}$. Conclude that for any $k : B \times A \longrightarrow C$ the transpose of $f \circ k$ is $f^A \circ \lambda k$.

Exercise 3.2.5. Show, for any object A and monic $f : B \longrightarrow C$, that f^A is monic.

3.3 Strict Initial Object

Definition 3.3.1 (Strict initial object)**.** In any category, an initial object is *strict* if and only if every object with a morphism to it is also initial.

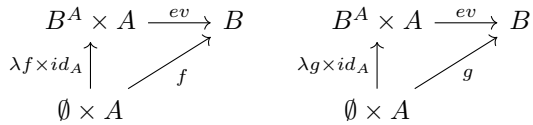
Initial objects in a Cartesian closed category are strict. This can be formally stated as the following theorem.

Theorem 3.3.2. In a Cartesian closed category with an initial object $\emptyset$, if an object A has a morphism $f : A \longrightarrow \emptyset$, then $A \cong \emptyset$.

Proof. See the rest of this section. □

Theorem 3.3.3. In a Cartesian closed category with an initial object $\emptyset$, $\emptyset \times A \cong \emptyset$ for every object A.

Proof. For any object B and any morphisms $f, g : \emptyset \times A \longrightarrow B$, the following diagrams commute.



However, since a morphism from $\emptyset$ to B is unique, $\lambda f = \lambda g$. Then,

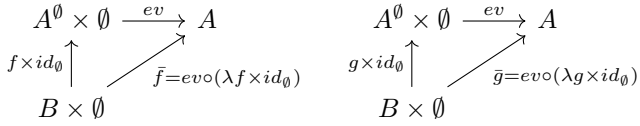
$$\begin{aligned} f &= ev \circ (\lambda f \times id_A) \\ &= ev \circ (\lambda g \times id_A) \\ &= g \end{aligned}$$

Hence there is only one morphism from $\emptyset \times A$ to B: that is, $\emptyset \times A$ is an initial object. By Theorem 2.1.5, $\emptyset \times A \cong \emptyset$. □

Remark 3.3.4. By Theorem 2.2.4 and Theorem 1.4.4, we also have $(A \times \emptyset) \cong (\emptyset \times A) \cong \emptyset$.

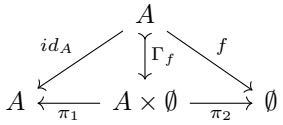
Theorem 3.3.5. In a Cartesian closed category with an initial object $\emptyset$, $A^{\emptyset} \cong 1$ for every object A.

Proof. For any object B and any morphisms $f, g : B \longrightarrow A^{\emptyset}$, the following diagrams commute.

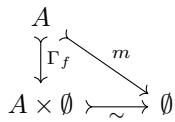


By Theorem 3.3.3, $B \times \emptyset$ is an initial object and there is only one morphism from $B \times \emptyset$ to A, namely, $\bar{f} = \bar{g}$. Then, $f = \lambda(\bar{f}) = \lambda(\bar{g}) = g$. This means that there is only one morphism from B to $A^{\emptyset}$, that is, $A^{\emptyset}$ is a terminal object. By Theorem 2.1.2, $A^{\emptyset} \cong 1$. □

Proof of Theorem 3.3.2. By Theorem 2.4.7 and Theorem 2.4.2, the graph of $f : A \longrightarrow \emptyset$ is monic.



By Theorem 3.3.3, Theorem 1.5.9 and Theorem 1.5.3, there is a monic $m : A \longrightarrow \emptyset$.



On the other hand, by Theorem 2.1.6, any morphism from A to $\emptyset$ is split epic. Therefore, by Theorem 1.5.9, m is iso and $A \cong \emptyset$. □

3.4 Partially-ordered Set

Definition 3.4.1 (Partially-ordered set)**.** A *partially-ordered set* $\langle L; \leq \rangle$ is a set L with the binary relation $\leq$ which is reflexive, anti-symmetric, and transitive.

Definition 3.4.2 (Poset)**.** A *poset* is a category in which, for any objects A and B, there is at most one morphism from A to B, and all the diagrams commute.

Theorem 3.4.3. A *poset* can be construed as a partially-ordered set L, whose elements are the objects of the poset, and for all $A, B \in L$, $x \leq y$ in L iff the poset has a morphism from x to y, and $x = y$ in L iff $x \cong y$.

Proof.

Reflexivity: For any object A, there exists id_A, thus $A \leq A$.

Anti-symmetry: For any object A and B with morphisms $f : A \longrightarrow B$ and $g : B \longrightarrow A$, $g \circ f = id_A$ since the morphism from A to A is unique, and $f \circ g = id_B$ since the morphism from B to B is unique. Thus $A \cong B$. Therefore, if $A \leq B$ and $B \leq A$, then $A = B$.

Transitivity: For any object A, B, and C with morphisms $f : A \longrightarrow B$ and $g : B \longrightarrow C$, there is a composite $g \circ f : A \longrightarrow C$. Therefore, if $A \leq B$ and $B \leq C$, then $A \leq C$.

□

Remark 3.4.4. Note that $A \leq B$ iff $A \times B \cong A$ (or $A + B \cong B$) in a poset.

Exercise 3.4.5. Show that a poset is a lattice iff it has binary products and coproducts. (Hint: products and coproducts are meets and joins in a poset.)

Exercise 3.4.6. Show that a poset is a bounded lattice iff it has a terminal object and an initial object. (Hint: a terminal object and an initial object are the top and the bottom elements.)

Exercise 3.4.7. Show that a poset is a Heyting lattice iff it is Cartesian closed. (Hint: exponentials are Heyting implications in a poset.)

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